

Physics-Informed Neural Networks for Scientific Computing

Bridging Physics and AI in
Groundwater Domain

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Dec 4, 2025



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de Strasbourg

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01

PINN Motivation

Why Combine Physics & Machine Learning?



01

Data-Driven Revolution:

Machine learning is now widely used in physics, finding patterns purely from data.

BUSINESS INSIDER

AI

Unstable genius: DeepMind cracks a century-old physics mystery with AI



By Alistair Barr

Author of the Tech Memo newsletter

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NEWS | 14 November 2025

A Chinese AI model taught itself basic physics – what discoveries could it make?



Home AI Data Center Driving Gaming Pro Graphics Rc

NVIDIA AI Physics Transforms Aerospace and Automotive Design, Accelerating Engineering by 500x

Industry leaders and software providers adopt NVIDIA DoMINO NIM microservice to advance computational engineering with AI-powered simulation and modeling.

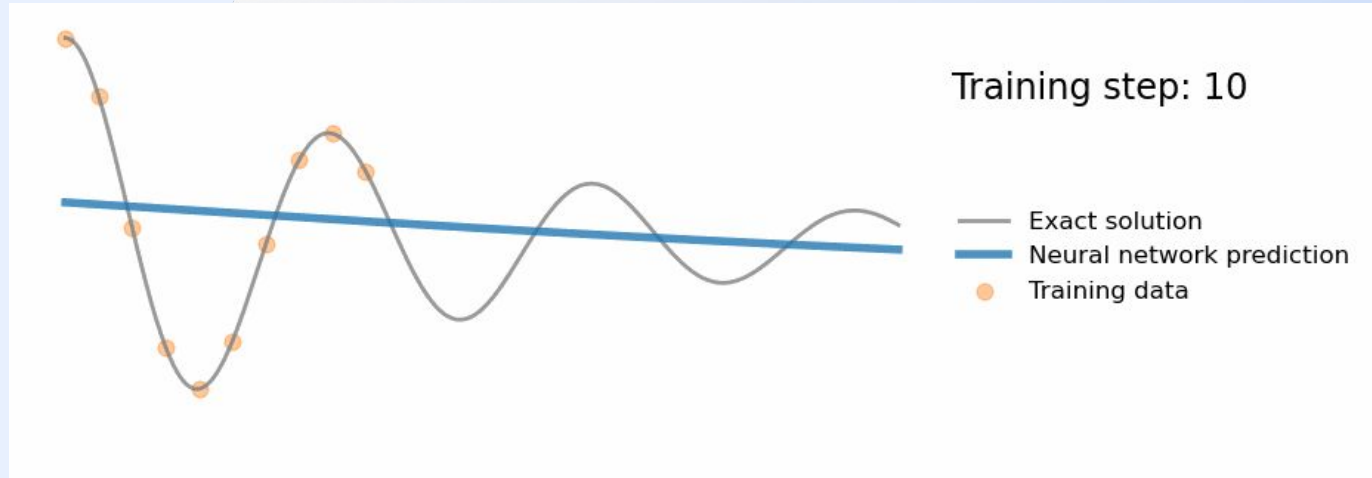
October 28, 2025 by [Timothy Costa](#)



02

Limitation:

Purely data-driven models can misbehave outside their training data – they lack understanding of the physics.

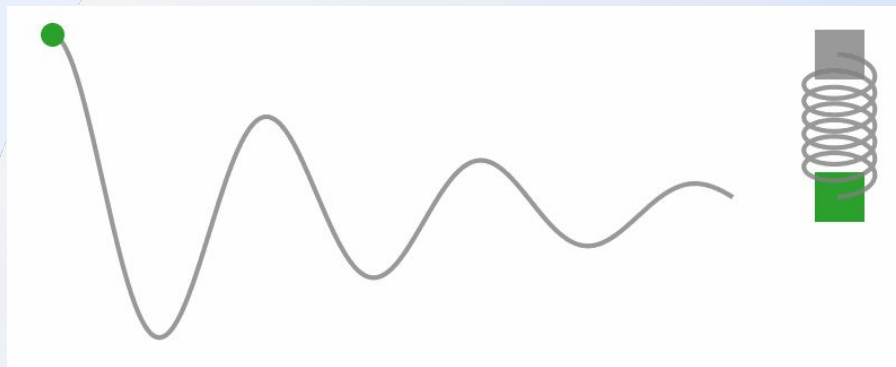


03

Scientific Knowledge:

We often do have well-established equations (PDEs/ODEs) from physics.

How can we leverage them?

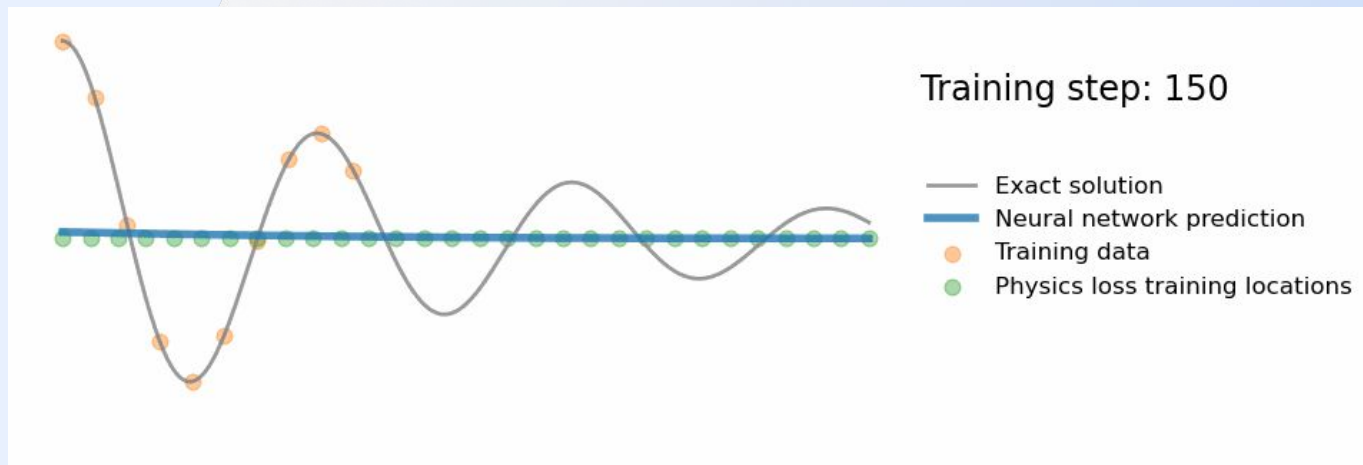


$$m \frac{d^2 u}{dx^2} + \mu \frac{du}{dx} + ku = 0$$

04

Enter SciML and PINNs:

The new paradigm is Scientific Machine Learning – use ML augmented with physics to get more reliable, data-efficient models.



02

PINN Theory & Example from Scratch

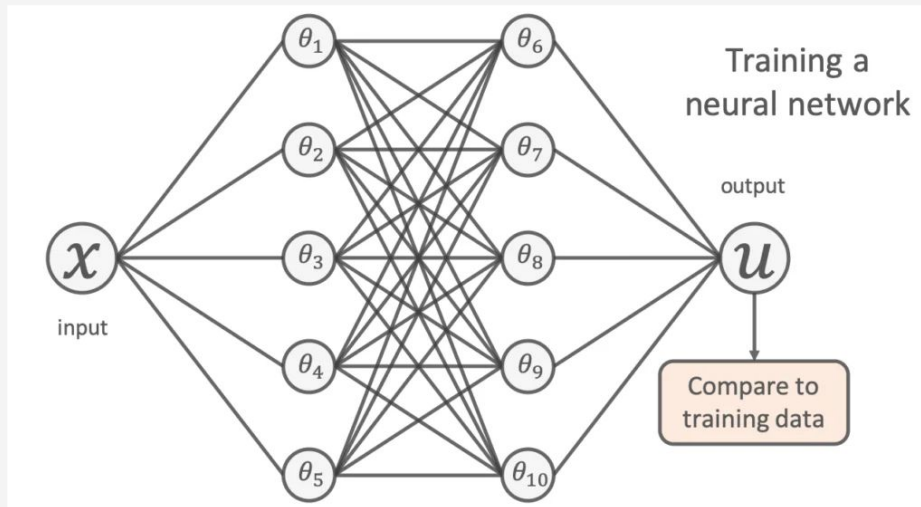


What is a Neural Network?

A model that learns input-output relations solely from data, and able to understand complex patterns.

Loss = Data Loss

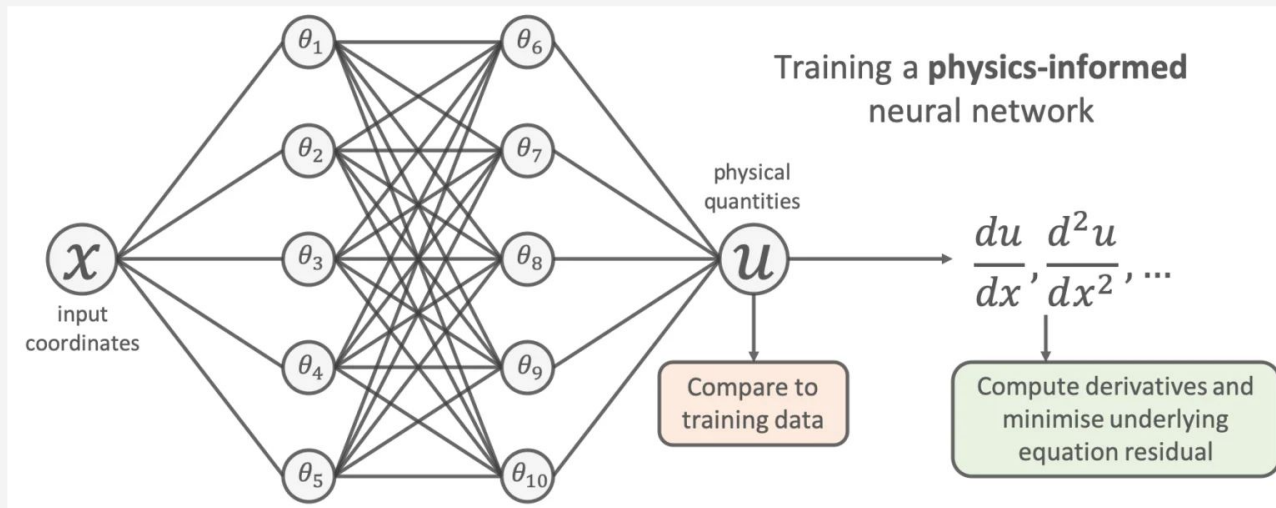
If the network's predictions mismatch the experimental data, it gets penalized.



$$\min \frac{1}{N} \sum_i^N (u_{\text{NN}}(x_i; \theta) - u_{\text{true}}(x_i))^2$$

What is a **Physics-Informed Neural Networks**?

a neural network
trained with physics
equations
embedded in its loss
function



Loss = Data Loss + Physics Loss

If the network also tries to violate the physics, it gets penalized – so it *learns solutions that satisfy the PDE*.

$$\min \frac{1}{N} \sum_i^N (u_{\text{NN}}(x_i; \theta) - u_{\text{true}}(x_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dx^2} + \mu \frac{d}{dx} + k \right] u_{\text{NN}}(x_j; \theta) \right)^2$$

How does a Physics-Informed Neural Networks work?

Hands-on example

$$y'(x) = -y(x)$$

$$y(0) = 1$$

$$y(x) = e^{-x}$$

A physics-informed neural network is a function $\hat{y}(x; \theta)$ with parameters θ (weights).

$$y'(x) = -y(x)$$

$$y(0) = 1$$

$$\hat{y}(x) = \dots$$

Step 1

Define Neural Network
Architecture

$$y'(x) = -y(x)$$

$$y(0) = 1$$

$$\hat{y}(x) = a_0 + a_1x + a_2x^2$$

$$a_0, a_1, a_2 = ?$$

Step 2

Define Loss: Observed
Data

Given data points: $(x_{\text{obs } i}, y_{\text{obs } i}), \quad i = 1, \dots, N$

PINN at $x_{\text{obs } i}$: $\hat{y}(x_{\text{obs } i}) = a_0 + a_1 x_{\text{obs } i} + a_2 x_{\text{obs } i}^2$

Data Residual: $r_{\text{data}}(x_{\text{obs } i}) = \hat{y}(x_{\text{obs } i}) - y_{\text{obs } i}$

$$\text{Data Loss: } L_{\text{data}} = \sum_{i=1}^N r_{\text{data}}(x_{\text{obs } i})^2$$

Step 3

Define Loss: PDE

PINN function: $\hat{y}(x) = a_0 + a_1x + a_2x^2$

By differentiation : $\hat{y}'(x) = a_1 + 2a_2x$

PDE Residual: $r_{\text{PDE}}(x) = \hat{y}'(x) + \hat{y}(x)$

PDE Loss:
$$L_{\text{PDE}} = \sum_{i=1}^N r_{\text{PDE}}(x_{\text{coll } i})^2$$

Step 4

Define Loss: Initial
Condition

Initial Condition: $y(0) = 1$

PINN at $x = 0$: $\hat{y}(0) = a_0$

IC Residual: $r_{\text{IC}} = \hat{y}(0) - 1$

IC Loss: $L_{\text{IC}} = r_{\text{IC}}^2 = (a_0 - 1)^2$

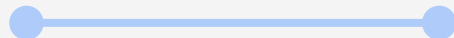
Step 5

Define Loss: Total Loss

Total Loss: $L_{\text{total}} = L_{\text{data}} + L_{\text{PDE}} + L_{\text{IC}}$

$$= \sum_{i=1}^N \left(r_{\text{data}}(x_{\text{obs } i}) \right)^2 + \sum_{j=1}^N \left(r_{\text{PDE}}(x_{\text{coll } j}) \right)^2 + r_{\text{IC}}^2$$

$$= \sum_{i=1}^N \left(\hat{y}(x_{\text{obs } i}) - y_{\text{obs } i} \right)^2 + \sum_{j=1}^N \left(\hat{y}'(x_{\text{coll } j}) + \hat{y}(x_{\text{coll } j}) \right)^2 + (\hat{y}(0) - 1)^2$$



Data Loss



Physics Loss

Step 6

Training

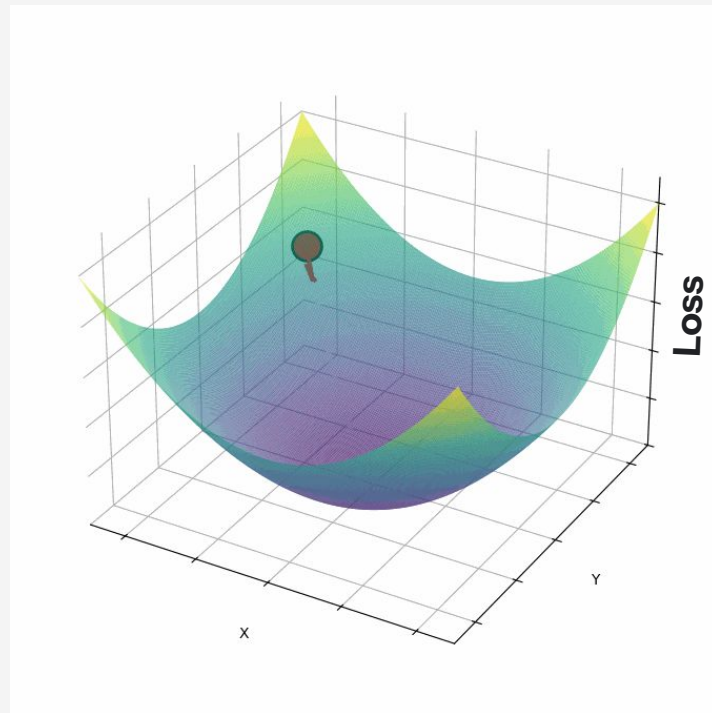
Total Loss: $L_{\text{total}} = L_{\text{data}} + L_{\text{PDE}} + L_{\text{IC}}$

$$= \sum_{i=1}^N (\hat{y}(x_{\text{obs } i}) - y_{\text{obs } i})^2 + \sum_{i=1}^N (\hat{y}'(x_{\text{coll } i}) + \hat{y}(x_{\text{coll } i}))^2 + (\hat{y}(0) - 1)^2$$

Gradients: $\frac{\partial L_{\text{total}}}{\partial a_0}, \frac{\partial L_{\text{total}}}{\partial a_1}, \frac{\partial L_{\text{total}}}{\partial a_2}$

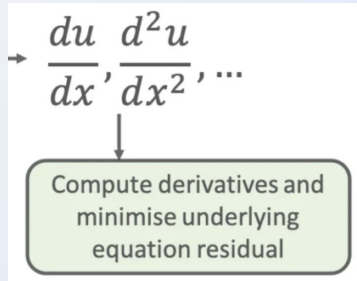
Update Rule: $a_k^{(n+1)} = a_k^{(n)} - \eta \frac{\partial L_{\text{total}}}{\partial a_k}$

Training Loop: $a_k^{(n+1)} = a_k^{(n)} - \eta \frac{\partial L_{\text{total}}}{\partial a_k}$ (repeat until convergence)



Under the Hood: Computing Derivatives

$$\min \frac{1}{N} \sum_i^N (u_{\text{NN}}(x_i; \theta) - u_{\text{true}}(x_i))^2 + \frac{1}{M} \sum_j^M \left(\left[m \frac{d^2}{dx^2} + \mu \frac{d}{dx} + k \right] u_{\text{NN}}(x_j; \theta) \right)^2$$



```
# compute the "physics loss"
```

```
yhp = model(x_physics)
```

```
dx = torch.autograd.grad(yhp, x_physics, torch.ones_like(yhp), create_graph=True)[0]
```

```
dx2 = torch.autograd.grad(dx, x_physics, torch.ones_like(dx), create_graph=True)[0]
```

```
physics = dx2 + mu*dx + k*yhp
```

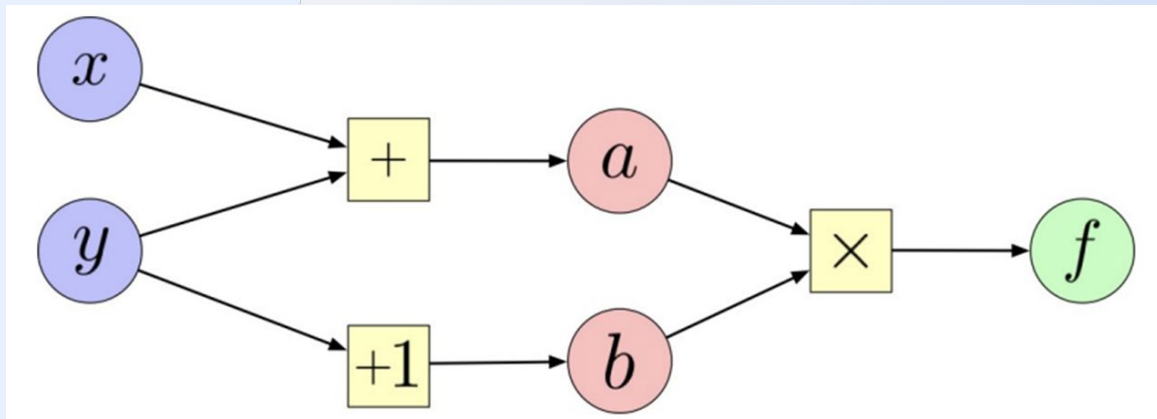
```
physics_loss = (1e-4) * torch.mean(physics**2)
```



Under the Hood: Automatic Differentiation

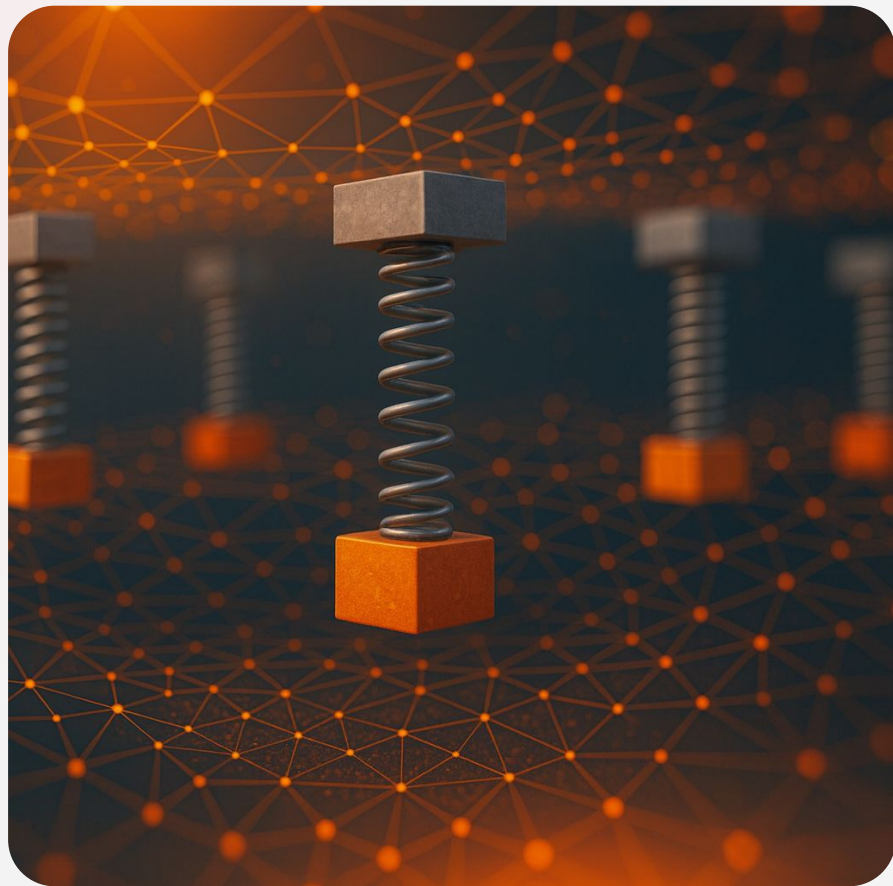
Automatic differentiation (AD) is a method for computing derivatives exactly and efficiently by systematically applying the chain rule to a model's operations.

As long as the model's architecture is end-to-end differentiable, automatic differentiation can be computed.



03

PINN Implementation & Demonstration



Damped Harmonic Oscillator

Governing PDE

$$m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt} + kx = 0$$

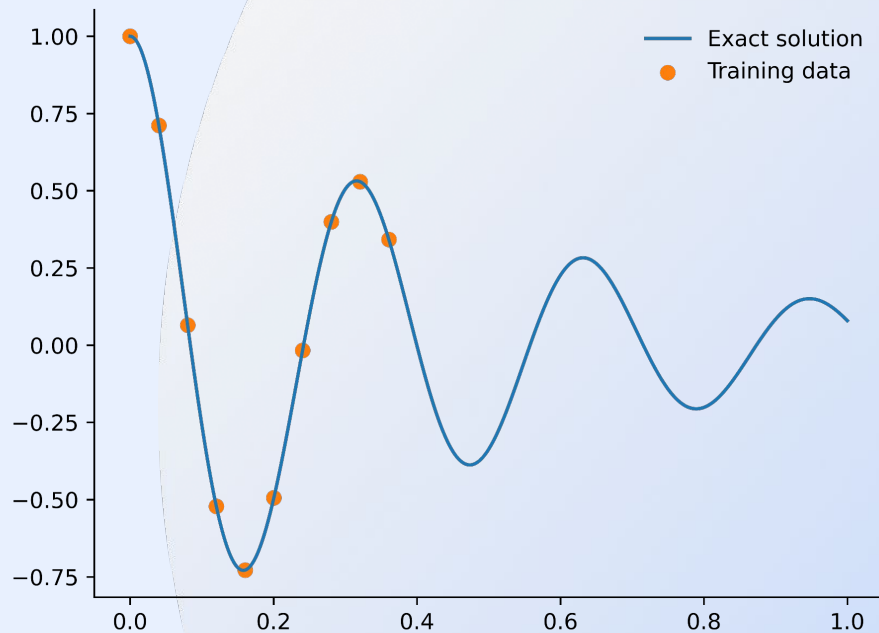
initial conditions

$$x(0) = 1, \quad \frac{dx}{dt} = 0$$

Exact Solution

$$x(t) = e^{-\delta t} (2A \cos(\phi + \omega t)), \quad \text{with } \omega = \sqrt{\omega_0^2 - \delta^2}$$

Exact solution and sample data points

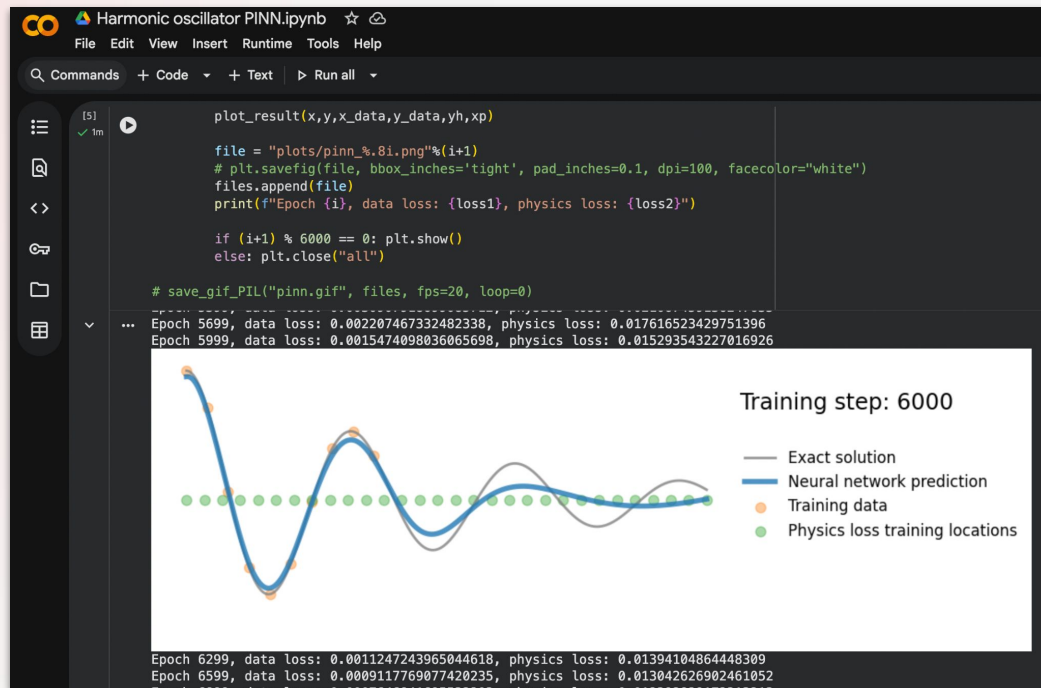


Network Architecture

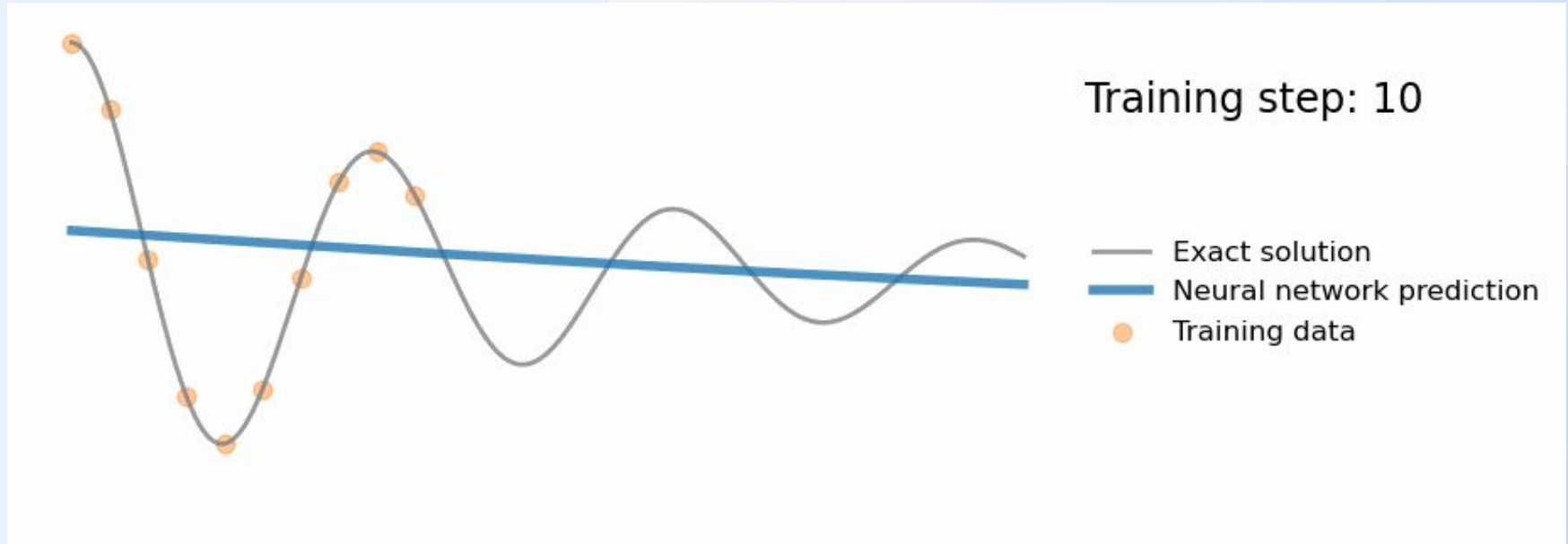
Fully Connected Network Architecture



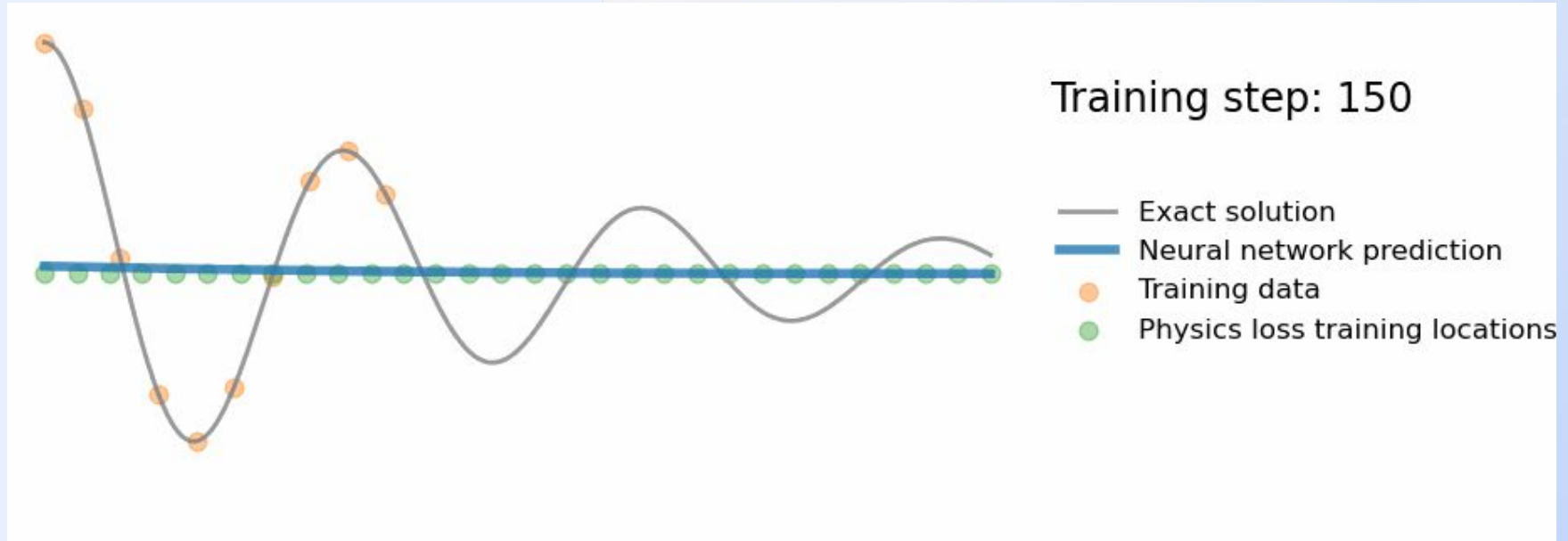
Live Demo



Training using a normal neural network

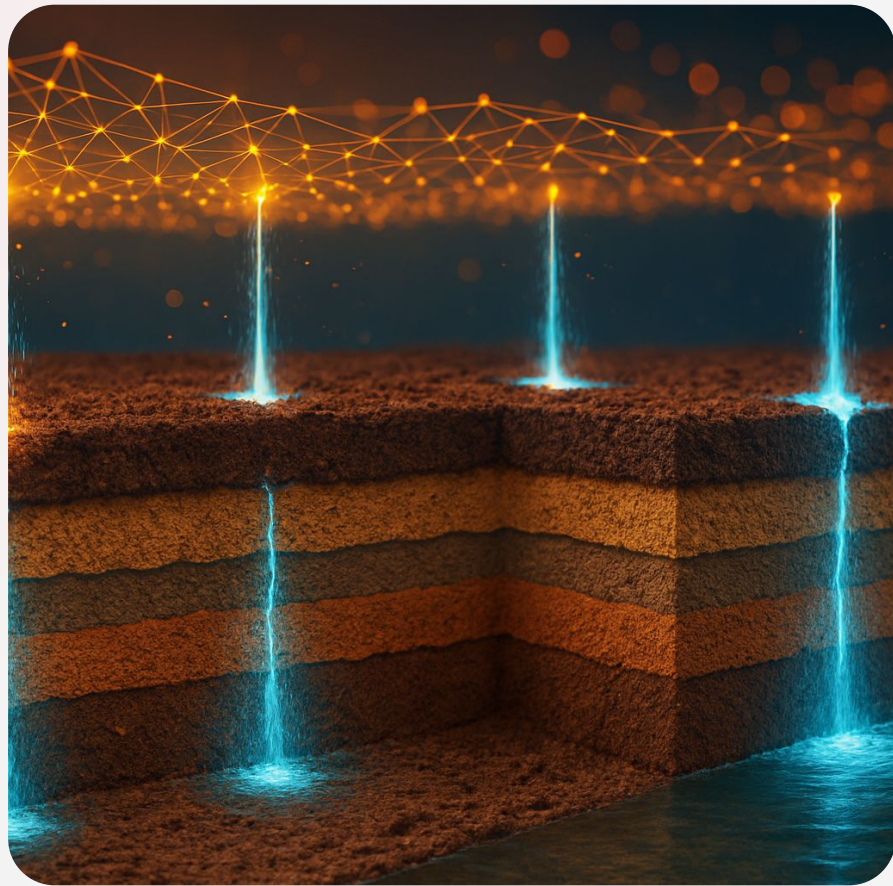


Training using a physics-informed neural network



04

PINNs for Groundwater Flow Modeling



Representative set of related research

2022

GW-PINN: A deep learning algorithm for solving groundwater flow equations

[link](#)

2023

Investigating steady unconfined groundwater flow using PINNS

[link](#)

2023

PINNs for solving transient unconfined aquifer flow

[link](#)

2024

PINNs in groundwater flow modeling: advantages and future directions

[review paper](#)

[link](#)

2025

PINNs for forward and inverse multispecies contaminant transport with variable parameters

[link](#)

2025

Hybrid Two-Stage Reconstruction of Multiscale Subsurface Flow with Physics-informed Residual Connected Neural Operator

[link](#)

05

Next Steps



Next Steps

- | | | |
|----|--|----------|
| 00 | Conduct literature review on PINNs in groundwater domain | Dec 2025 |
| 01 | Decide on problem & case study | Jan 2026 |
| 02 | Define thesis scope & focus | Feb 2026 |
| 03 | Curate data, design architectures, craft loss functions, & optimize | 1H 2026 |
| 04 | Conduct sensitivity and comparative analyses | Q2 2026 |
| 05 | Write the research manuscripts | Q3 2026 |

References

Raissi et al., 2019

PINN **Founding Paper** ✨ ✨ ✨

Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations.

[link](#)

Ben Moseley, 2021

Code basis for live demo

So, What Is a Physics-Informed Neural Network?

[link](#)

Brunton, 2024

Helpful video teaching PINNs

Physics Informed Neural Networks (PINNs)

[link](#)

Belforion.de

Physics behind live demo

Damped Harmonic Oscillator - Derivation and Solution of the Differential Equations.

[link](#)

Thank you



Slidedeck available at kutt.it/pinn

